Implementation of the social force model



Macroscopic model validation

The pedestrian ‘fundamental diagram’ is the relationship between crowd density and crowd flow (or average speed), where:

- Crowd density ρ (m^{-2}): Number of people per square metre.
- Crowd speed v (m/s): Average walking speed for a uni-directional flow of people.
- Crowd flow Φ ($m^{-1}s^{-1}$): Number of people crossing a line per second, per metre.
- We also have: $\Phi = \rho v$

Empirical density-flow data

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The fundamental diagram of pedestrian movement revisited

Armin Seyfried[1], Bernhard Steffen[1], Wolfram Klingsch[2] and Maik Boltes[1]

[1] Central Institute for Applied Mathematics, Research Centre Jülich, 52425 Jülich, Germany
[2] Institute for Building Material Technology and Fire Safety Science, University of Wuppertal, Pauluskirchstrasse 7, 42285 Wuppertal, Germany
E-mail: A.Seyfried@fz-juelich.de, B.Steffen@fz-juelich.de, klingsch@uni-wuppertal.de and M.Boltes@fz-juelich.de

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Empirical density-flow data

Experimental setup:

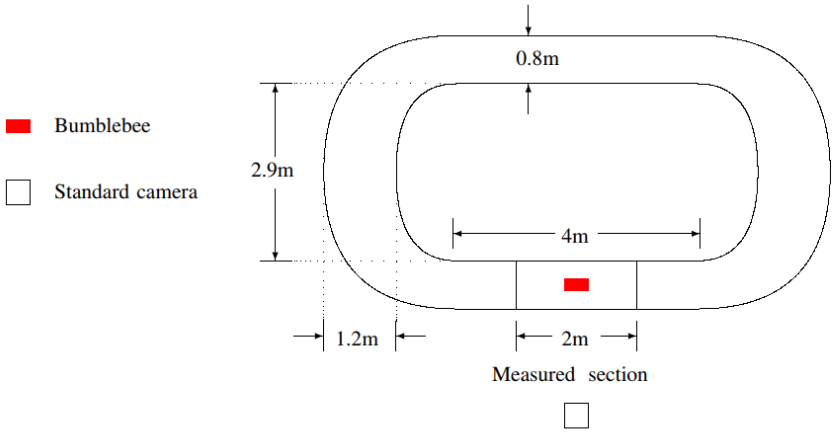


Figure 2. Experimental set-up for the measurement of the velocity–density relation for the single-file movement.

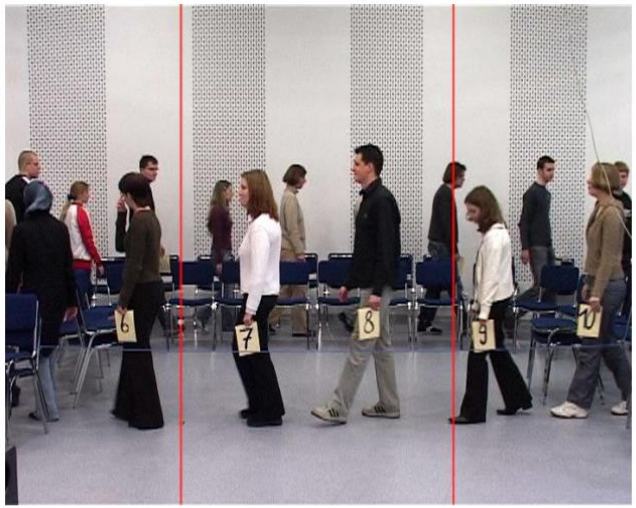


Figure 3. One frame of the standard video recording of the cycle with $N = 20$ after the preparation for the manual analysis.

Empirical density-flow data

Experimental results:

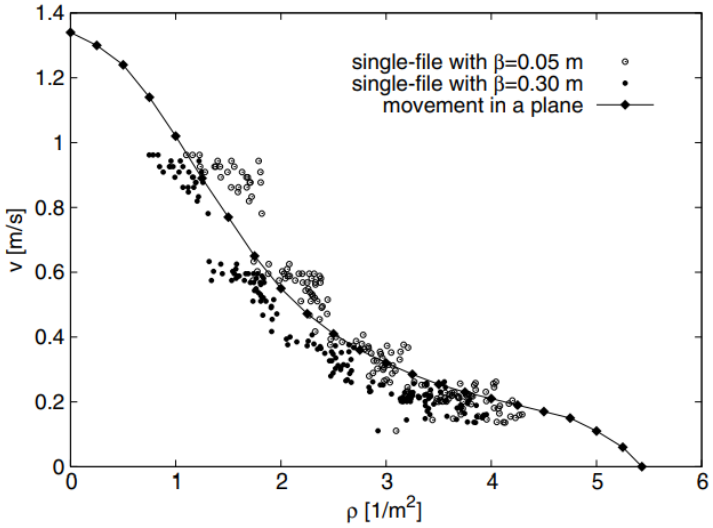
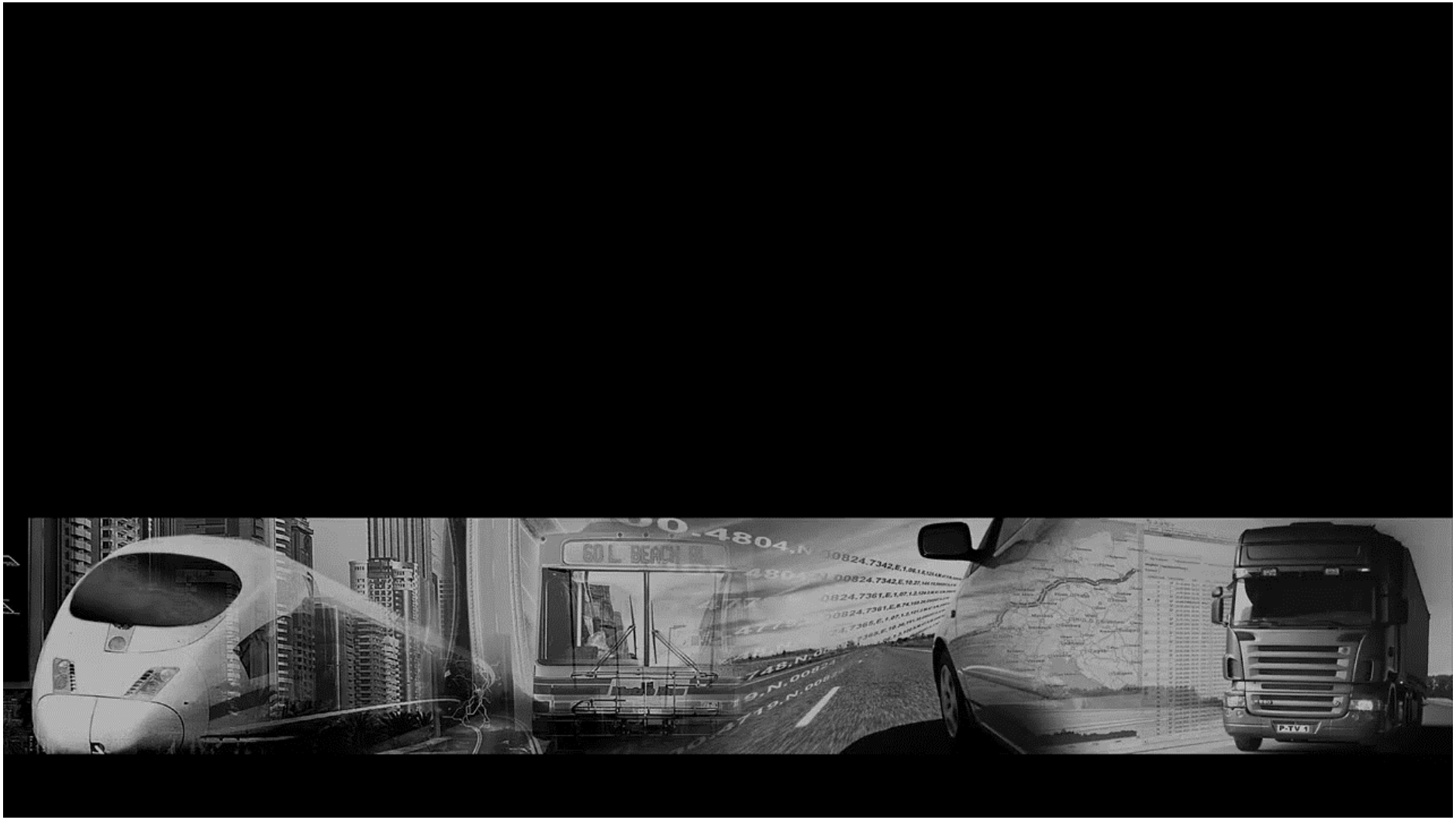


Figure 8. Comparison of the velocity–density relation for the single-file movement with the movement in a plane according to Weidmann [25]. For the scaling of the line-density we choose $\beta = 0.05$ s and $\beta = 0.3$ s; see equation (2).

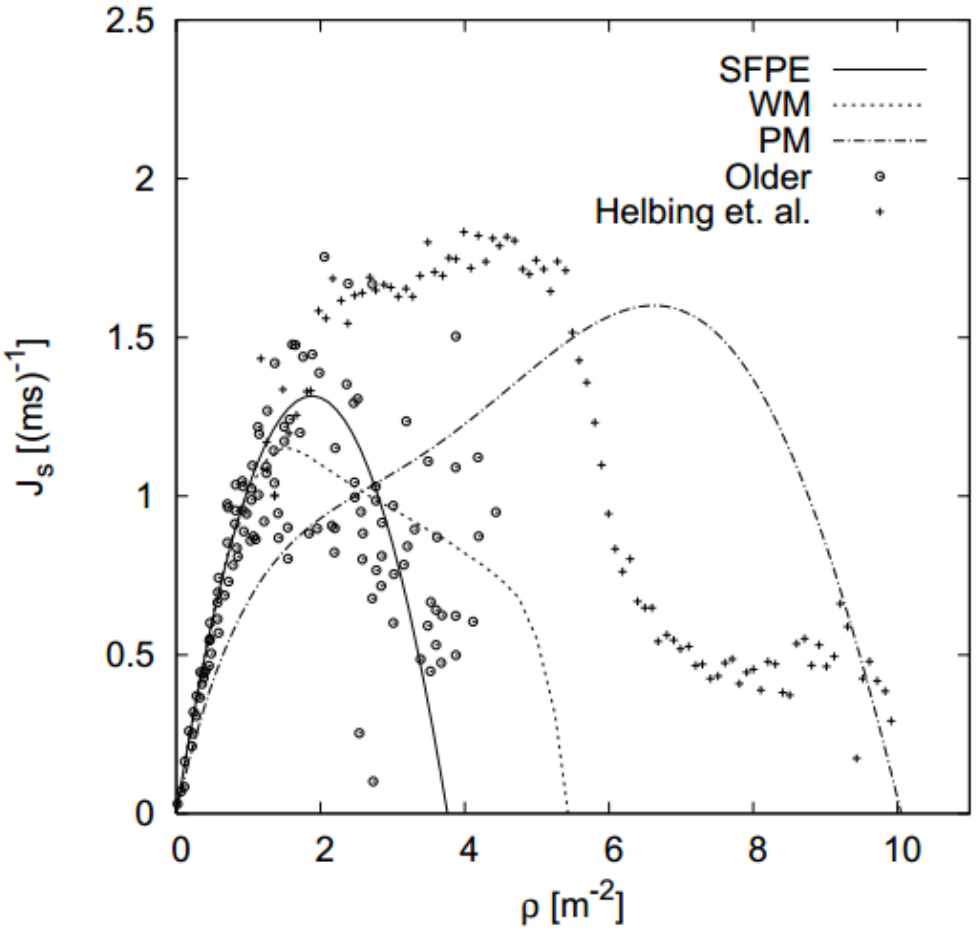
Table 1. Comparison of mean values and standard deviations (σ) gained from the automated and manual procedures.

N	ρ (1 m^{-1})	ρ^{man} (1 m^{-1})	v^{man} (m s^{-1})	v^{aut} (m s^{-1})
1			1.24 (0.15)	1.37 (0.21)
15	0.87	0.77 (0.12)	0.90 (0.05)	0.88 (0.15)
20	1.16	1.07 (0.11)	0.56 (0.05)	0.51 (0.11)
25	1.45	1.39 (0.12)	0.35 (0.04)	0.26 (0.14)
30	1.73	1.71 (0.17)	0.23 (0.03)	0.16 (0.15)
34	1.97	1.76 (0.24)	0.17 (0.03)	0.13 (0.34)


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Macroscopic model validation

Empirical data:



Schadschnedier et al (2008) Evacuation Dynamics: Empirical Results, Modeling and Applications, Arxiv preprint.

Empirical micro data collection

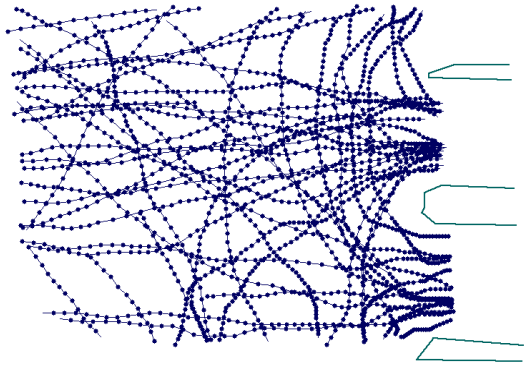
Data from different videos are used:

A Johansson, D Helbing, and P S Shukla (2007) Specification of the social force pedestrian model by evolutionary adjustment to video tracking data. Advances in Complex Systems 10, 271-288

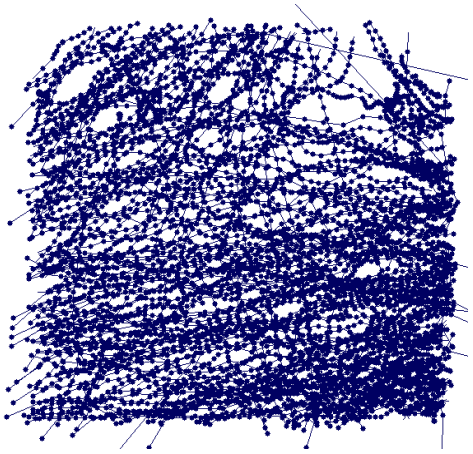




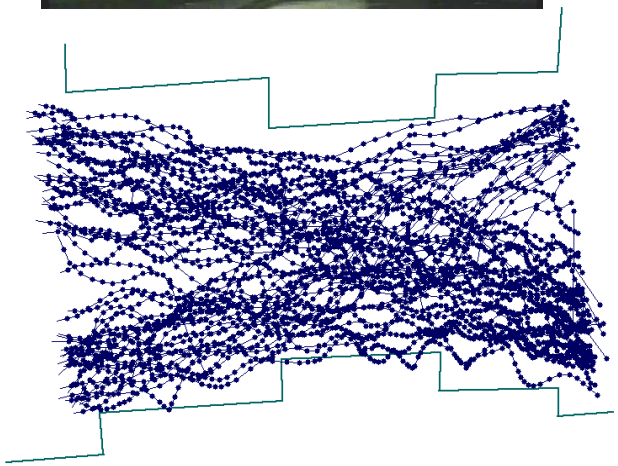




Budapest, Hungary



Budapest, Hungary



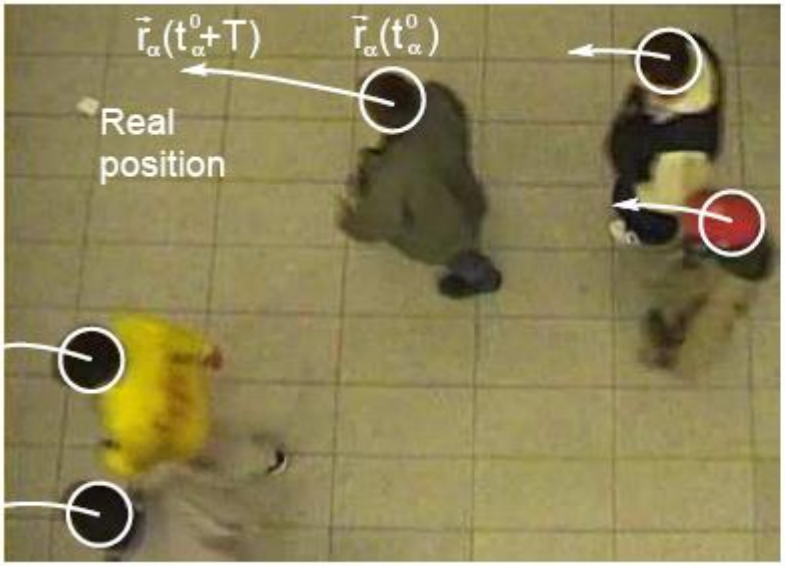
Dresden, Germany

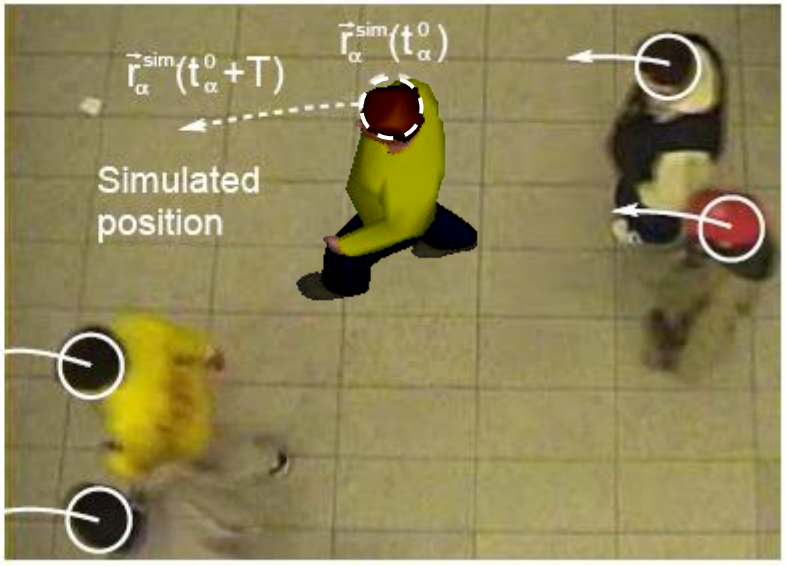
Micro calibration and validation

Replace: A pedestrian from the empirical trajectory data is replaced by a simulated pedestrian.

Measure: The deviation of the simulated position to the position in the data gives an error measure.

Optimize: This error measure is used by an optimization procedure in order to find model parameters that minimize the error.



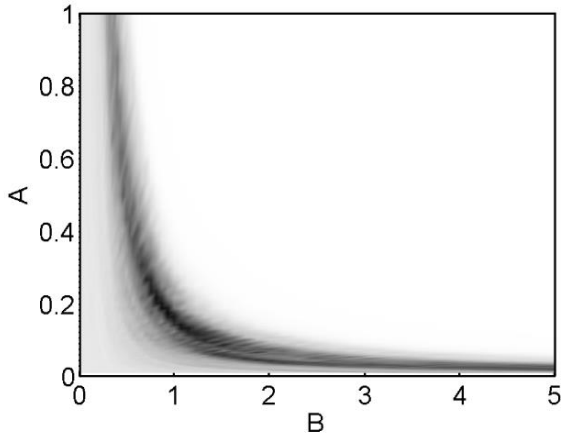


Microscopic model validation & calibration

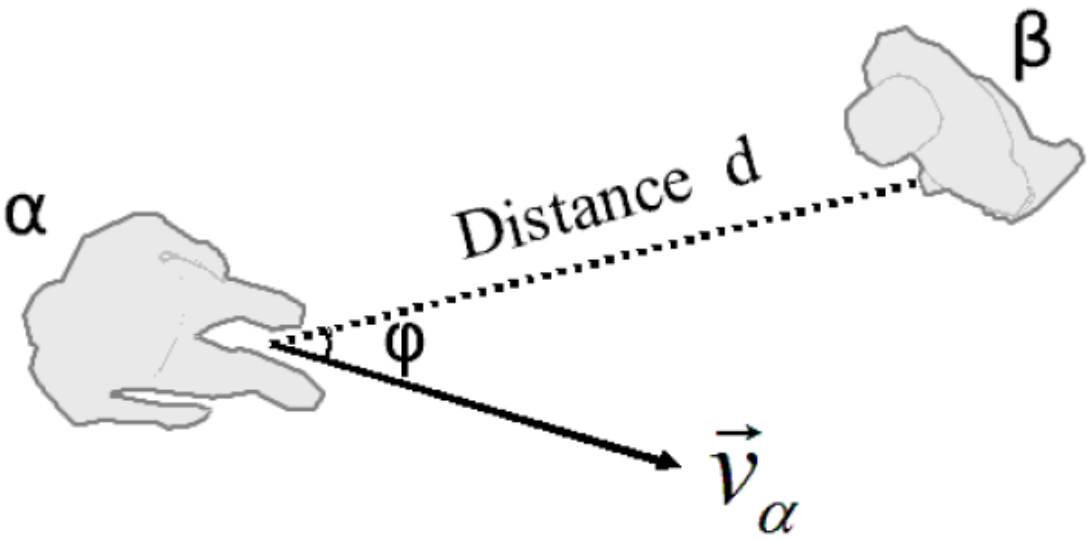
For a number of model specifications, optimization processes are carried out in order to find the optimum parameters and goodness of fit values.

$$\vec{f}_{\alpha\beta}(t) = A_s \frac{\vec{d}}{\left\|\vec{d}\right\|} \exp(-\left\|\vec{d}\right\| / B_s)$$

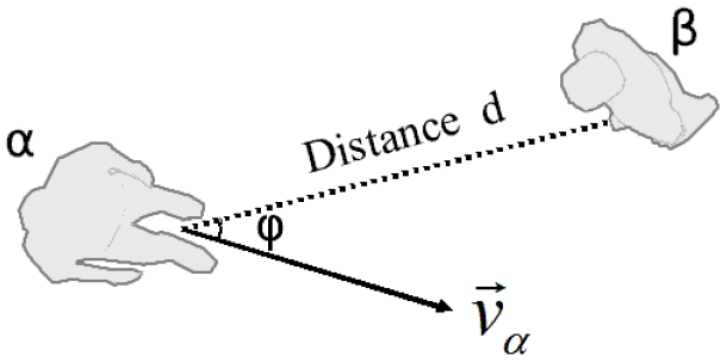
Model	N	Angular dependence	A_s	B_s	τ	Speed percentile	ΔT
Circular	17	λ =0.02	0.04	1.28	0.59	48	-
Elliptical I	9	λ =0.22	0.85	0.38	0.64	53	0.87
Elliptical II	9	λ =0.25	0.42	0.48	0.61	51	1.04
Circular	4	$\lambda = 1$	0.03	2.30	0.57	43	-
Elliptical I	1	$\lambda = 1$	0.52	0.37	0.56	42	0.53
Elliptical II	13	$\lambda = 1$	0.39	0.31	0.56	43	1.88
Circular	10	Half circle	0.79	0.37	0.61	52	-
Elliptical I	19	Half circle	0.24	0.63	0.61	53	1.54
Elliptical II	10	Half circle	0.39	0.54	0.61	53	0.47
Circular	∞	λ =0.14	0.03	2.03	0.62	52	-
Elliptical I	∞	λ =0.11	0.23	0.45	0.61	43	2.26
Elliptical II	∞	λ =0.09	0.05	1.47	0.63	54	1.17
Circular	∞	$\lambda = 1$	0.70	0.26	0.61	42	-
Elliptical I	∞	$\lambda = 1$	0.03	0.96	0.57	42	1.00
Elliptical II	∞	$\lambda = 1$	0.55	0.32	0.56	43	0.91
Circular	∞	Half circle	0.11	1.05	0.62	53	-
Elliptical I	∞	Half circle	0.60	0.40	0.60	53	1.62
Elliptical II	∞	Half circle	2.27	0.18	0.53	43	0.20



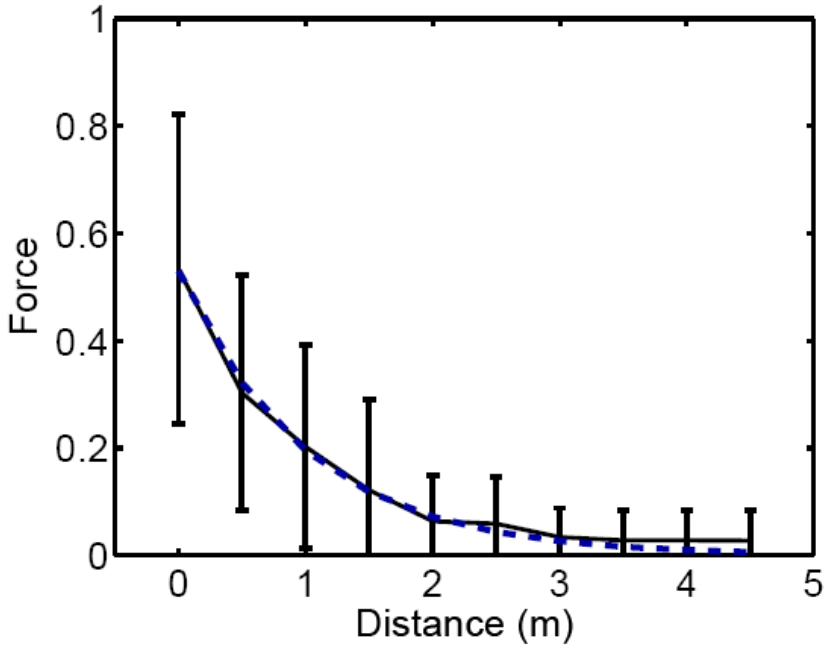
Social force calibration



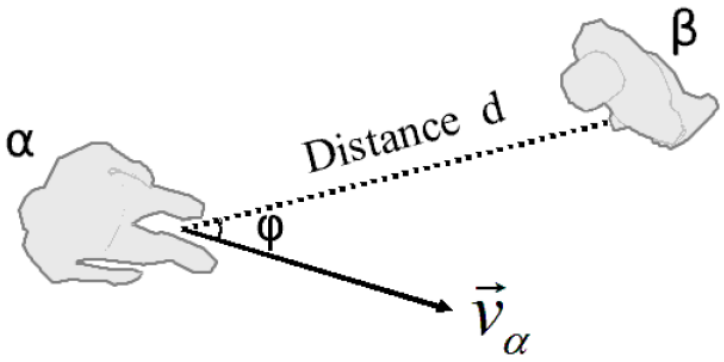
Social force calibration



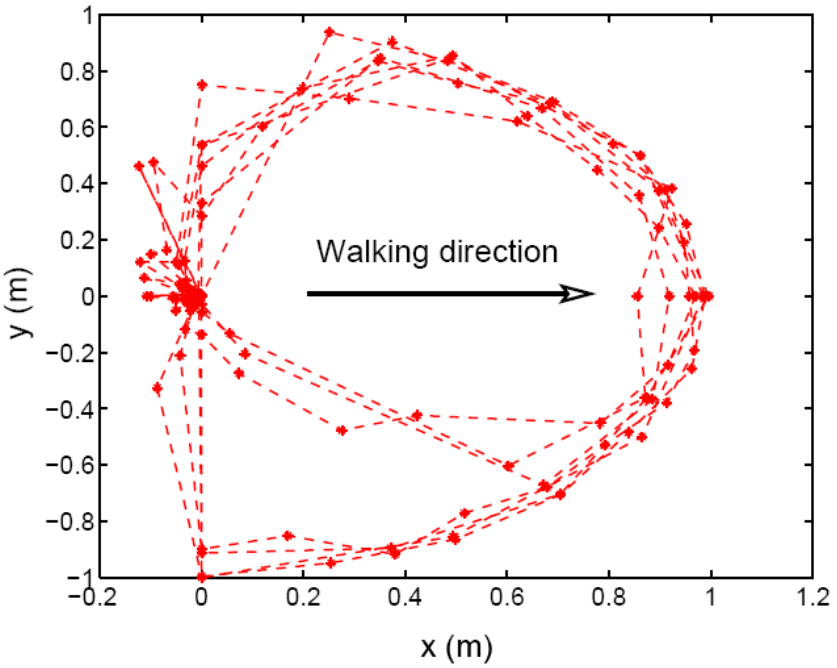
Distance dependence



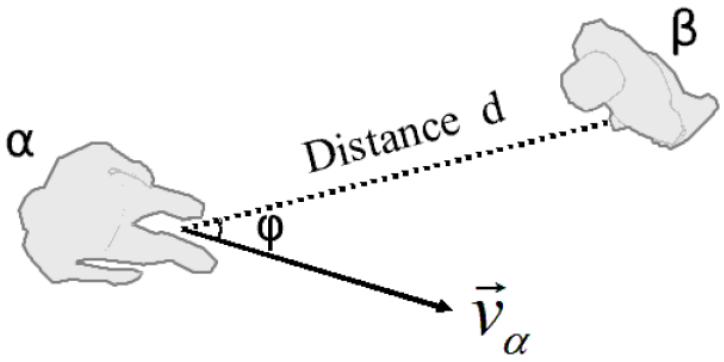
Social force calibration



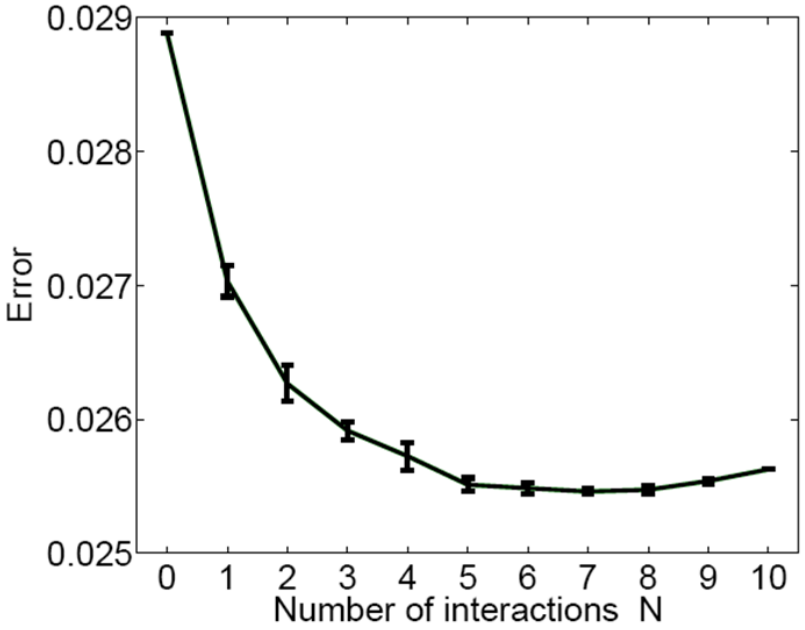
Angular dependence



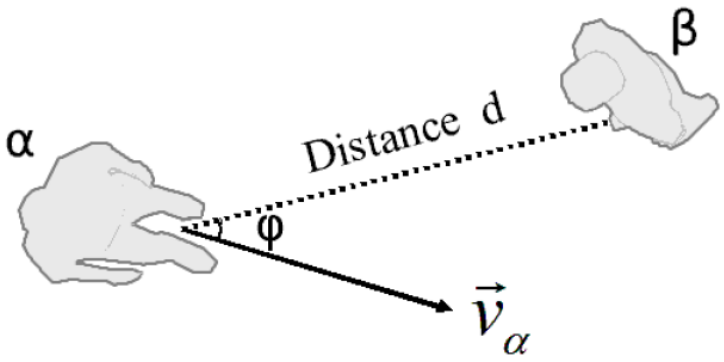
Social force calibration



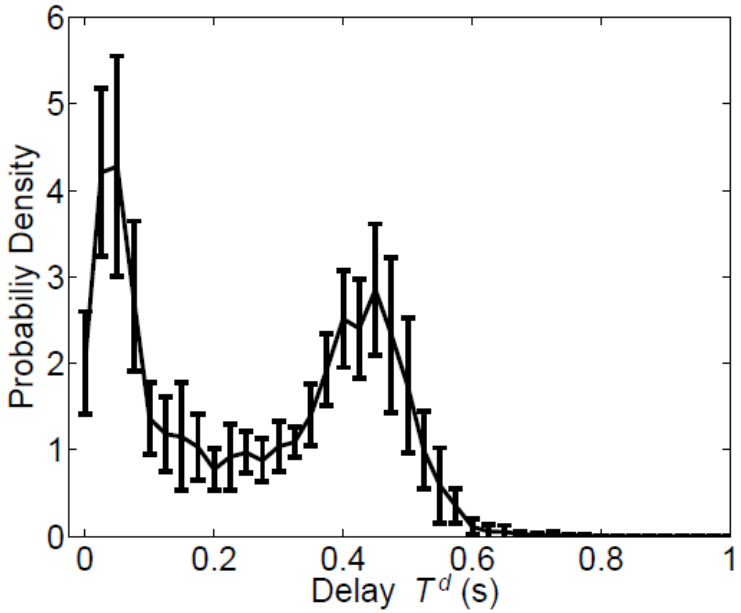
Number of interactions



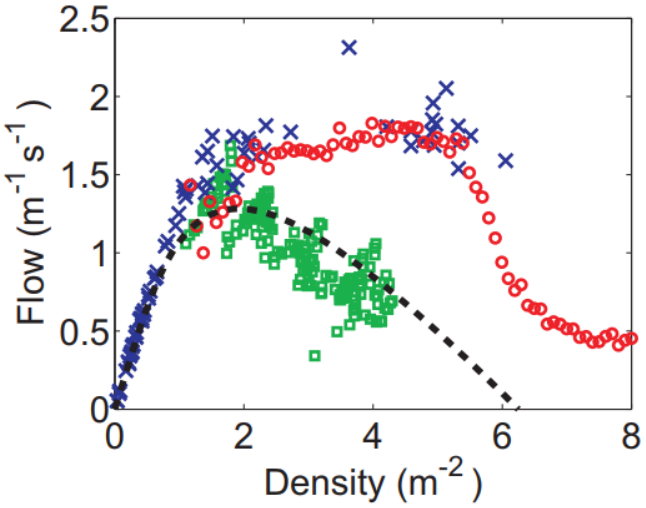
Social force calibration



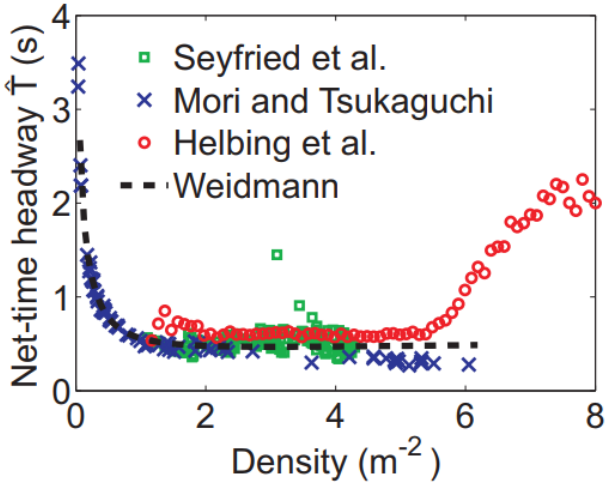
Delayed response



Net-time headway



$$v(\varrho) = \frac{d - 2r}{\hat{T}} = \frac{1/\sqrt{\varrho} - 1/\sqrt{\varrho_{max}}}{\hat{T}}$$



Johansson (2009), PRE

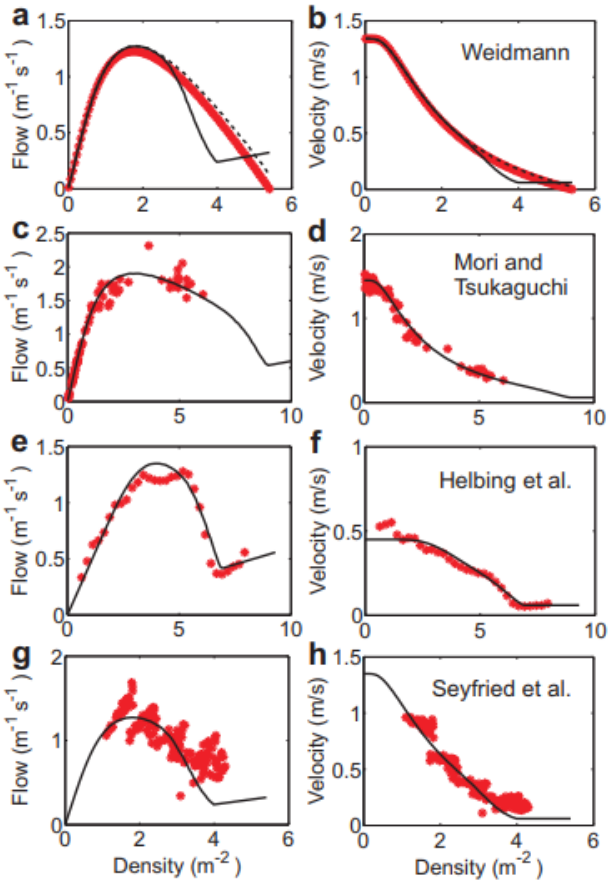


Figure 7: (Color online) Fundamental diagrams and velocity-density relations generated by Eqs. (1) to (3), assuming a constant net-time headway $\hat{T} = 0.5$ s and a minimum velocity $v_{min} = 0.06$ m/s. Red markers represent empirical data and solid lines the theoretically expected relationships. a) Flow-density data, and b) velocity-density data by Weidmann [3], compared to a fundamental diagram generated with the parameters $\varrho_{max} = 5.4$ m^{-2} and $v_{max} = 1.34$ m/s. The dashed line shows the result when it is assumed that no pedestrians are stopping, i.e. $f_{stop} = 0$. c) Flow-density data, and d) velocity-density data by Mori and Tsukaguchi [4], compared to a fundamental diagram generated with the parameters $\varrho_{max} = 12$ m^{-2} and $v_{max} = 1.45$ m/s. e) Flow-density data, and f) velocity-density data from Helbing *et al.* [2], compared to a fundamental diagram generated with the parameters $\varrho_{max} = 9.3$ m^{-2} and $v_{max} = 0.45$ m/s. g) Flow-density data, and h) velocity-density data from Seyfried *et al.* [11], compared to a fundamental diagram generated with the parameters $\varrho_{max} = 5.4$ m^{-2} and $v_{max} = 1.35$ m/s.